Introduction : Scope of extractive metallurgy, occurrence of metals in nature, minerals and ores. Elementary concepts of Mineral processing and extraction of metals (5 L)

Pyrometallurgy: Drying and calcinations, roasting & derivation of roasting conditions by Kelloss's diagram , relevance of Ellingham diagram in metal extraction, reduction of metal oxides, matte smelting and converting, metal refining processes: fire-refining, liquation and distillation. (13 L)

Hydrometallurgy: Leaching and its methods, construction & use of Paurbax diagram, bioleaching, solution purification and concentration: solvent extraction and ion exchange. Recovery of metals from leach solutions. (7 L)

Electrometallurgy : Principles of electrolysis, electrolytic systems, electro-refining, electrowinning and other electro-metallurgical processes. (4 L)

Process Flow Sheets: Production of iron and steel, aluminium, copper, zinc and lead. (6 L)

Analysis of unit processes: Reactor kinetics, heat and material balance. (3 L)

4. READINGS

4.1 TEXTBOOKS::

1.J. Newton: Extractive Metallurgy, Wiley.
3. W.H. Dennis: Extractive Metallurgy- Principles an Applications, Pitman.
4. J.D. Gilchrist: Extraction Metallurgy, Pergamon.
5. R.D. Pehlke: Unit Processes in Extractive Metallurgy, Elsevier.
6. T. Rosenqvist: Principles of Extractive Metallurgy, McGraw Hill.
7. C.B. Gill: Nonferrous Extractive Metallurgy, Wiley-Interscience.
8. H.S. Ray and A. Ghosh: Principles of Extractive Metallurgy, New Age International Publishers.
9. H.S.Ray, R. Sridhar and K.P. Abraham: Extraction of Non-ferrous Metals, Affiliated East West.
10. F. Habbashi, Principles of Extractive Metallurgy, Vol 1-4, McGraw-Hill